

Live Product Audit + UX / Brand Redesign Brief

What is live now, what is broken, and the target consumer experience.

VERSION	STATUS	TARGET
v0.3	Implementation brief	OpenCode / implementation team

Executive decision: Treat the current Vercel build as a technical prototype, not a product to polish screen-by-screen. The next branch should be an EmOcean AI v2 experience with a persistent camera, randomized repeated trials, full-screen stimuli, an engaging baseline, realistic chat-theme testing, and a genuinely useful result/export.

P0 before visual polish: Fix real iPhone Safari camera reliability. If the camera does not run on the target phone, the product cannot be evaluated meaningfully.

1. What the product should be

EmOcean AI is a privacy-first browser experience that measures observable physiological and facial responses while a person experiences controlled changes in digital environments. The first consumer use case is to discover which colors and chat themes are associated with the user's calmest and most positively activating response.

The promise is not "AI knows your emotion." The promise is closer to: **your body responds to your digital environment; EmOcean helps you explore that response.**

Recommended product structure

Stage	Consumer idea	Technical purpose
Meet your signals	See what your camera can observe	Camera + rPPG + expression + motion + SQL
Find your baseline	Establish your normal session reference	Within-user normalization
Color Currents	Experience randomized colors	Repeated controlled color trials
Chat Tides	See the winning colors in a real chat	Theme-level confirmation
Your EmOcean	Get a personal result and export	Ranking, uncertainty, wallpaper/theme export

2. Current live-build issues to fix

Issue	Why it matters	Priority
Brand still reads like a private expression / mirror prototype	Does not explain the color-personalization payoff or the reason to spend several minutes	P0
iOS camera failure on at least one real phone	Blocks the core experience on a major target platform	P0
Camera not continuously visible	Users do not understand what is being measured and lose trust during waiting/calibration	P0
Long linear 30-second color exposures	Weak consumer pacing and weaker experimental control because order, drift, fatigue and condition are confounded	P0
Small color presentation area	The stimulus is too weak visually and does not resemble the environment the user will later use	P0
Look-away/rest instruction	Wrong for a screen-response study; user should look comfortably at the screen	P0
Paced/slow-breathing cues during baseline	Changes autonomic physiology and contaminates the baseline	P0
Live pseudo-state scores such as calm/focus/curious	Can imply more validated scientific precision than the sensing model supports	P1
Chat experience is not a real chat-theme experiment	Does not test the actual background/bubble environment that users will export	P1
Waiting feels idle	Users may abandon before the useful part begins	P1

3. Brand and visual direction

Consumer name: **EmOcean AI**. Domain target: **emocean.ai**. The identity should feel oceanic, fluid, playful and premium without becoming psychedelic, clinical or generic SaaS.

Recommended core copy

"Your screen affects you."

"Let's find out how."

Supporting copy: **EmOcean AI uses your camera to observe subtle changes in your pulse, breathing and facial response while you experience different digital environments.**

Primary CTA: **Find my colors** Secondary CTA: **How it works**

Trust chips: No wearable | Runs on your device | Video never uploaded | About 5-7 minutes

Brand palette

Name	HEX	Use
Deep Ocean	#103438	Primary dark / text / header
Lagoon	#1CB7A1	Primary accent
Aqua	#61D6E5	Secondary accent / data
Coral	#FF7D73	Warm action / contrast
Sun	#FFC85A	Highlight / progress
Lavender	#A894F4	Secondary emotional accent
Foam	#F6FBF9	Background

Important: Brand colors are not the experimental color set. Experimental colors must be controlled separately, ideally with matched perceptual lightness.

4. Landing page copy and flow

Hero

Your screen affects you.
Let's find out how.

EmOcean AI uses your camera to observe subtle changes in your pulse, breathing and facial response while you experience different digital environments.

CTA: Find my colors

How it works

1	Meet your signals	See your pulse, breathing estimate, motion and signal quality from a normal camera.
2	Experience the currents	Colors and digital environments appear in a randomized repeated sequence.
3	Compare your response	We compare each condition with your own session baseline.
4	Take it with you	Get a background, color recipe and chat-theme preview to recreate in everyday apps.

Privacy line

Your camera is analyzed on this device. Raw video is not recorded or uploaded. Results are for personal exploration and are not medical measurements.

5. Camera consent, setup and calibration

The camera should become visible immediately after permission and remain visible until the experiment ends. On desktop use a small floating mirrored preview. On mobile use a compact floating preview that does not obscure instructions.

Consent screen copy

Camera privacy Your camera is analyzed on this device to estimate physiological and facial-response signals. Raw video is not recorded or uploaded. Closing this page ends camera access. Results are for personal exploration and are not medical measurements.

Calibration instructions

Look comfortably at the screen. Breathe normally. Blink normally. Try not to talk or move too much.

Do not ask the user to look away. Do not pace breathing. Slow breathing itself changes cardiac/autonomic measures and would contaminate the baseline.

Make the wait entertaining without changing physiology

- Keep the mirrored camera visible.
- Show a soft ocean-ripple progress animation.
- Advance through neutral status messages every few seconds: Finding your face -> Checking the light -> Finding your pulse -> Learning your normal rhythm -> Almost there.
- Show signal quality and gentle corrective guidance such as "Move toward softer front light" or "Hold still for a moment."
- Do not use intense motion, emotional imagery or breathing coaching during baseline.

6. Experimental UX: colors should fill the screen

Remove the small color-card treatment. The trial stimulus should dominate the visual field. On mobile the screen should feel like a real app environment; the camera preview and quality indicator float above it.

During trials, hide exact HR / breathing numbers. Showing the live numbers creates biofeedback and can make users consciously change their state. Show only: Measuring | Signal good | Hold still | Improve lighting.

Consumer quick protocol

Stage	Duration	Design
Baseline	25-30 s	Neutral screen, natural breathing
Color screening A	5 colors x 12 s	Random order; about 4 s neutral washout
Color screening B	5 colors x 12 s	New random order; same conditions repeated
Finalists	Top 2 x about 30 s	Randomized confirmation with neutral washout
Chat-theme stage	Short repeated trials	Realistic chat UI using finalist palettes
Subjective choice	About 10 s	Ask what felt best only after physiology trials
Results	No fixed limit	Calm Current, Activation Current, confidence, export

Analysis window: Mark the first about 3 seconds after each visual transition as transition time and exclude it from the primary response estimate. Reject low-quality windows instead of forcing a result.

7. Chat Tides: make it a real chat experience

The chat stage should resemble a polished messaging application, not a rating survey with color swatches. A trial theme should include background, incoming bubble, outgoing bubble, text color, accent and timestamps. Bubble colors must remain readable.

Neutral chat examples

"Did you leave the notebook on the second shelf?"

"Yes, beside the blue folder."

"What time does the cafe open?"

"Eight, according to the sign."

Create multiple balanced neutral message sets with similar word count, difficulty, number of bubbles and display timing. Randomize message-set assignment independently of the theme condition so content does not systematically confound color.

Separate variables before combining them

- Color Currents: background color only; all other visual variables fixed.
- Chat Tides: test complete chat themes derived from the best color candidates.
- Typography: test fonts separately on a constant color before combining finalist font + color.
- Emoji Mirror: separate stage; keep color and font fixed while emoji changes.
- Finalist combination: only after individual effects are understood.

8. Measurement UI: show what is defensible

Before the experiment, use the sensing stack to create the wow moment. The public live screen should prioritize measured or transparently derived quantities.

Signal	Recommended display
Heart rate	BPM + quality
Breathing	breaths/min estimate + confidence
Pulse waveform	simple animated trace
Pulse variability	only when enough clean beats exist; describe as camera-derived pulse variability / PRV
Facial response	positive / neutral / negative tendency or calibrated expression probabilities
Motion / head pose	stable / moving; face centered / away
Signal quality	percentage or qualitative grade

Avoid: Do not make unvalidated "Calm 0.28", "Focus 0.63" or "Curious 0.19" scores the main scientific output unless a specific validated model exists for those quantities.

9. Results that are useful and shareable

The result should feel like a personalized digital identity, not a lab report.

YOUR CALM CURRENT

SEA GLASS

#BBD3C6

Compared with your session baseline: heart rate -4.1 BPM; breathing -1.6/min; movement -12%; repeated response 2/2; confidence 84%.

Then show a full chat preview plus: [Download background](#) | [Copy colors](#) | [WhatsApp setup](#) | [Telegram setup](#).

Subjective versus physiological result

Memorable moment: You chose Lavender. Your strongest calm-associated physiological response was Sea Glass.

Only show a stress-like result when several independent signals agree. Higher heart rate by itself is activation, not proof of stress.

10. iOS / mobile acceptance requirements

- Request camera from an explicit user tap.
- Use HTTPS and navigator.mediaDevices.getUserMedia.
- Prefer facingMode: "user" without an unnecessarily strict exact constraint.
- Use video autoplay + muted + playsInline and attach stream via srcObject.
- Keep the same video element and CameraProvider mounted across experiment stages.
- Prefer one persistent /experience route/state machine rather than route changes that remount the camera.
- Handle visibilitychange/pageshow and MediaStreamTrack ended/mute/unmute recovery.
- Use 100dvh, safe-area insets and real-device testing on iPhone Safari.
- Provide readable error states for permission denied, missing camera, model-load failure, insufficient light and stalled stream.

Target state machine

```
CONSENT -> CAMERA_SETUP -> SIGNAL_ACQUISITION -> BASELINE -> COLOR_A -> COLOR_B -> FINALISTS -> CHAT_THEME ->  
OPTIONAL_FONT -> OPTIONAL_EMOJI -> SUBJECTIVE_CHOICE -> RESULTS -> EXPORT
```

11. Tooling recommendation for this existing build

Because this project already exists as code and is deployed on Vercel, use **OpenCode for implementation/refactoring** and optionally **v0 for focused visual redesign concepts**. Do not migrate the project merely to obtain a new website builder shell.

Tool	Use here	Do not use it for
OpenCode	Repository inspection, camera lifecycle, sensing integration, experiment engine, tests, refactors	Pure visual ideation without code context
v0	High-end UI concepts and visual component redesign in the existing repo workflow	Replacing scientific/sensing logic just to make the UI pretty
Lovable	Still strong for greenfield visual building and built-in browser testing	Forcing a migration from an already-running repo when unnecessary

12. Primary reference links

Current live deployment: <https://emo-ocean-neon.vercel.app>

Vital Camera SDK: <https://github.com/KegangWangCCNU/vitalcamera-sdk>

FacePhys browser demo: <https://github.com/KegangWangCCNU/FacePhys-Demo>

MediaPipe Face Landmarker - Web: https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker/web_js

MediaPipe web setup / browser support: https://ai.google.dev/edge/mediapipe/solutions/setup_web

MDN getUserMedia: <https://developer.mozilla.org/en-US/docs/Web/API/MediaDevices/getUserMedia>

OpenCode docs: <https://opencode.ai/docs/>

v0 GitHub docs: <https://v0.dev/docs/github>

v0 Design Mode: <https://v0.dev/docs/design-mode>

Lovable FAQ / GitHub limitations: <https://docs.lovable.dev/introduction/faq>

Lovable browser testing: <https://docs.lovable.dev/features/browser-testing>

Bottom line: The product should feel like an entertaining personal discovery experience, but the sensing and experiment rules must remain stricter than the visual layer. The fastest path is to fix the iPhone camera and experiment state machine first, then layer the new EmOcean AI design over reliable measurement.